CSA5117

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CRYPTOGRAPHY & NETWORK SECURITY

ASSIGNMENT NO : 3

Set ①

Consider the following two hash algorithms : MD5 and SHA-1. Both are used to produce message digests.

a) Compare the Output lengths of MD5 and SHA-1.
b) Explain why MD5 is no longer Considered Secure.
c) If two different files result in the Same hash in MD5, what type of attack does this represent?
d) Why is SHA-256 Considered more Secure than SHA-1? Illustrate with reference to Collision resistance.

ANSWER

* MD5, SHA-1 and SHA-256 Hash Algorithms

INTRODUCTION:

A Cryptographic hash algorithm Converts data of any Size into a fixed-length value Called a message digest or hash value. Hash functions are used for data integrity, authentication, digital Signatures, and file verification.

a) Output Length of MD5 and SHA-1

MD5 (Message-Digest Algorithm 5) produces a 128-bit (16-byte) message digest, represented by 32 hexadecimal characters.

SHA-1 (Secure Hash Algorithm 1) produces a 160-bit (20-byte) message digest, represented by 40 hexadecimal characters.

Feature	MD5	SHA-1
Output Length	128 bits	160 bits
Output in bytes	16 bytes	20 bytes
Hexadecimal characters	32	40
Security	Broken	Broken

b) Why MD5 is No Longer Secure

MD5 is no longer considered secure because it has serious collision vulnerabilities.

A collision occurs when two different inputs produce the same hash:

$$\text{Message 1} \neq \text{Message 2} \quad \text{but}$$

$$MD5(\text{Message 1}) = MD5(\text{Message 2})$$

Attackers can deliberately create different files with the same MD5 harsh. this can allow malicious files to appear identical to legitimate files when MD5 is used for verification.

MD5 should therefore not be used for:

* Digital Signatures
* Certificates
* Security-critical file verification.
* Cryptographic authentication.

It may still be used for accidental file corruption, but not against intentional attackers.

c) Same MD5 Hash for Two Different Files

If two different files produce the same MD5 hash, it is Called a 'hash Collision'

For Example:

File A ≠ File B

MD5 (File A) = ABC123

MD5 (File B) = ABC123

If an attacker deliberately Creates Such files, it is Called a Collision attack.

This is dangerous because an attacker Could replace a geniune file with a malicious file having the Same hash

d) Why SHA-256 is More Secure Than SHA-1

SHA-256 belongs to the SHA-2 family and Produces a 256-bit hash, while SHA-1 Produces a 160-bit harsh.

A larger hash provides greater Collision resistance.

For an ideal hash function, the approximate Complexity of a Collision attack is:

* SHA-1 : 2^{80} operations
* SHA-256 : 2^{128} operations

Thus, SHA-256 provides much Stronger protection against Collision attacks.

SHA-1 has already been demonstrated to have Practical Collision attacks, including the well-known SHAttered attack, where researchers produced two different with the Same SHA-1 hash.

No practical collision attack is currently known against SHA-256.

Comparision:

Algorithm	Hash Size	Collision Resistance	Status
MD5	128 bits	$\sim 2^{64}$	Not Secure
SHA-1	160 bits	$\sim 2^{80}$	Not Secure
SHA-256	256 bits	$\sim 2^{128}$	Secure

CONCLUSION:

MD5 produces a '128-bit hash' and 'SHA-1' produces a '160-bit hash.' Both are no longer recommended for Security-Sensitive applications because of their Collision weaknesses. A Situation where two different files have the Same MD5 hash is Called 'Collision.'

∴ SHA-256 is more Secure than SHA-1 because of its larger 256-bit output and much Stronger Collision resistance.